MLPerf® Storage v2.0 System Description

curvine_c5n9xlarge_fuse_1c_2s_16h

Curvine v0.2.0 (commit 77320b86) distributed file system, FUSE 3 client, OpenDAL-S3 UFS. 1 AWS c5n.9xlarge FUSE client and 2 AWS i3en.12xlarge Curvine worker nodes; the Curvine master is co-located on a worker node. In this round the Curvine worker service was limited to 16 vCPU / 123 GiB. 16 simulated NVIDIA H100 accelerators (DLIO). MLPerf Storage v2.0 closed division, training benchmark.

Submission

Submitter	Curvine
System name	curvine_c5n9xlarge_fuse_1c_2s_16h
Submission date	2026-10-28
Benchmark version	MLPerf Storage v2.0
Solution type	shared file
Submission division	CLOSED

System Configuration

Storage location	remote
Client software	proprietary
Storage interface	file
Required rack units	N/A
Multi-host support	True
Simultaneous write support	True
Simultaneous read support	True
Max sequential read	N/A
Max sequential write	N/A
Use local storage	False
Clients with integrated storage	0
Architecture	shared file (Curvine FUSE over AWS S3 with worker NVMe cache)

Node: dlio_client

Role	dlio_client
Quantity	1
Model	AWS EC2 c5n.9xlarge
CPU	1 x Intel Xeon Platinum 8124M @ 3.00GHz (18 cores)
Memory	96 GiB
PSU	N/A x N/A W, N/A
Network	data: AWS Elastic Network Adapter (ENA) 50Gbps x1
OS	Rocky Linux 9 (kernel 5.14.x, x86_64)
MPI	Open MPI 4.1+; env: N/A; launch flags: --oversubscribe, --allow-run-as-root (cosmoflow and 8c runs only)

Node: storage_data_node

Role	storage_data_node
Quantity	2
Model	AWS EC2 i3en.12xlarge
CPU	2 x Intel Xeon Platinum 8175M @ 2.50GHz (24 cores)

Memory	384 GiB
PSU	N/A x N/A W, N/A
Network	data: AWS Elastic Network Adapter (ENA) 50Gbps x1
OS	Rocky Linux 9 (kernel 5.14.x, x86_64)
MPI	N/A; env: N/A

Power Requirements

Role	Quantity	PSU nameplate (W)	Design power (W)	Active/Passive PSUs
dlio_client	1	N/A / N/A / N/A	N/A	N/A / N/A
storage_data_node	2	N/A / N/A / N/A	N/A	N/A / N/A

Notes

• Storage software: Curvine v0.2.0 (commit 77320b86), FUSE 3 client mounted on each DLIO client, OpenDAL-S3 as the underlying file system (UFS). Data are stored on the local NVMe devices of the i3en.12xlarge worker nodes; the Curvine master is co-located on one worker node.
• In this round the Curvine worker service was limited to 16 vCPU / 123 GiB per node, whereas a full i3en.12xlarge provides 48 vCPU / 384 GiB; this is the main difference to the Alluxio reference configuration.
• Simulated H100 accelerator counts: 16 (resnet50), 1 (unet3d), 5 (cosmoflow).
• Client CPU, core count and memory are taken from the run artifacts (summary.json): Intel Xeon Platinum 8124M, 18 cores / 36 threads, 91.78 GiB available out of 96 GiB.
• OS release, kernel and MPI version are the values declared at test time; they were not captured from the run hosts.
• Accelerators are simulated by the MLPerf Storage reference benchmark (DLIO); no physical GPU was used.
• Datagen was executed once per dataset before the measured runs; five consecutive runs were measured per model.
• AWS does not publish per-instance power data, therefore the power fields are N/A.

Appendix: system_configuration YAML (curvine_c5n9xlarge_fuse_1c_2s_16h.yaml; long lines are folded to the page width)

```
System:
  name: curvine_c5n9xlarge_fuse_1c_2s_16h
  description: "Curvine v0.2.0 (commit 77320b86) distributed file system, FUSE 3 client, OpenDAL-S3 UFS. 1 AWS c5n.9xlarge
FUSE client and 2 AWS i3en.12xlarge Curvine worker nodes; the Curvine master is co-located on a worker node. In this round
the Curvine worker service was limited to 16 vCPU / 123 GiB. 16 simulated NVIDIA H100 accelerators (DLIO). MLPerf Storage
v2.0 closed division, training benchmark."
  storage_location: remote
  client_software: proprietary
  storage_interface: file
  required_rack_units: N/A
  shared_capabilities:
    multi_host_support: True
    simultaneous_write_support: True
    simultaneous_read__support: True
  max_sequential_read: N/A
  max_sequential_write: N/A
  use_local_storage: False
  num_clients_with_integrated_storage: 0
  system_architecture: shared file (Curvine FUSE over AWS S3 with worker NVMe cache)

PowerRequirements:
  dlio_client:
    quantity: 1
    psu1_nameplate_power: N/A
    psu2_nameplate_power: N/A
    psu3_nameplate_power: N/A
    design_power: N/A
    num_active_psus: N/A
    num_passive_psus: N/A
  storage_data_node:
    quantity: 2
    psu1_nameplate_power: N/A
    psu2_nameplate_power: N/A
    psu3_nameplate_power: N/A
    design_power: N/A
    num_active_psus: N/A
    num_passive_psus: N/A

Nodes:
  dlio_client:
    quantity: 1
```

```
    hardware:
      model: AWS EC2 c5n.9xlarge
      rack_units: N/A
      power_supplies: N/A
      psu_configuration: N/A
      psu_rating: N/A
      memory_capacity: 96 GiB
      memory_configuration: N/A
      cpu_qty: 1
      cpu_model: Intel Xeon Platinum 8124M @ 3.00GHz
      cpu_cores: 18
    networking:
      data:
        model: AWS Elastic Network Adapter (ENA)
        speed: 50Gbps
        qty: 1
    operating_system:
      name: Rocky Linux
      version: "9"
      release_date: N/A
      kernel_version: "5.14.x"
      cpu_architecture: x86_64
    tuning:
      mpi_configuration:
        environment_variables: ["N/A"]
        version: Open MPI 4.1+
        launch_flags: ["--oversubscribe", "--allow-run-as-root (cosmoflow and 8c runs only)"]
      sysctl_parameters:
        - fs.file-max=1048576
        - net.core.rmem_max=134217728
        - net.core.somaxconn=65535
        - vm.max_map_count=2621440
        - vm.swappiness=1
      fuse_configuration:
        mount_point: /curvine-fuse
        direct_io: true
        attr_timeout_ms: 86400000
        entry_timeout_ms: 86400000
        node_cache_ttl: 24h
        enable_meta_cache: true
        meta_cache_timeout: 24h
        max_readahead_kb: 1024
        read_chunk_size: 1m
        read_chunk_num: 16
        client_storage_type: MEM
  storage_data_node:
    quantity: 2
    hardware:
      model: AWS EC2 i3en.12xlarge
      rack_units: N/A
      power_supplies: N/A
      psu_configuration: N/A
      psu_rating: N/A
      memory_capacity: 384 GiB
      memory_configuration: N/A
      cpu_qty: 2
      cpu_model: Intel Xeon Platinum 8175M @ 2.50GHz
      cpu_cores: 24
    networking:
      data:
        model: AWS Elastic Network Adapter (ENA)
        speed: 50Gbps
        qty: 1
    operating_system:
      name: Rocky Linux
      version: "9"
      release_date: N/A
      kernel_version: "5.14.x"
      cpu_architecture: x86_64
    tuning:
      mpi_configuration:
        environment_variables: ["N/A"]
        version: "N/A"
      sysctl_parameters:
        - fs.file-max=1048576
        - net.core.rmem_max=134217728
        - net.core.somaxconn=65535
        - vm.max_map_count=2621440
        - vm.swappiness=1

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